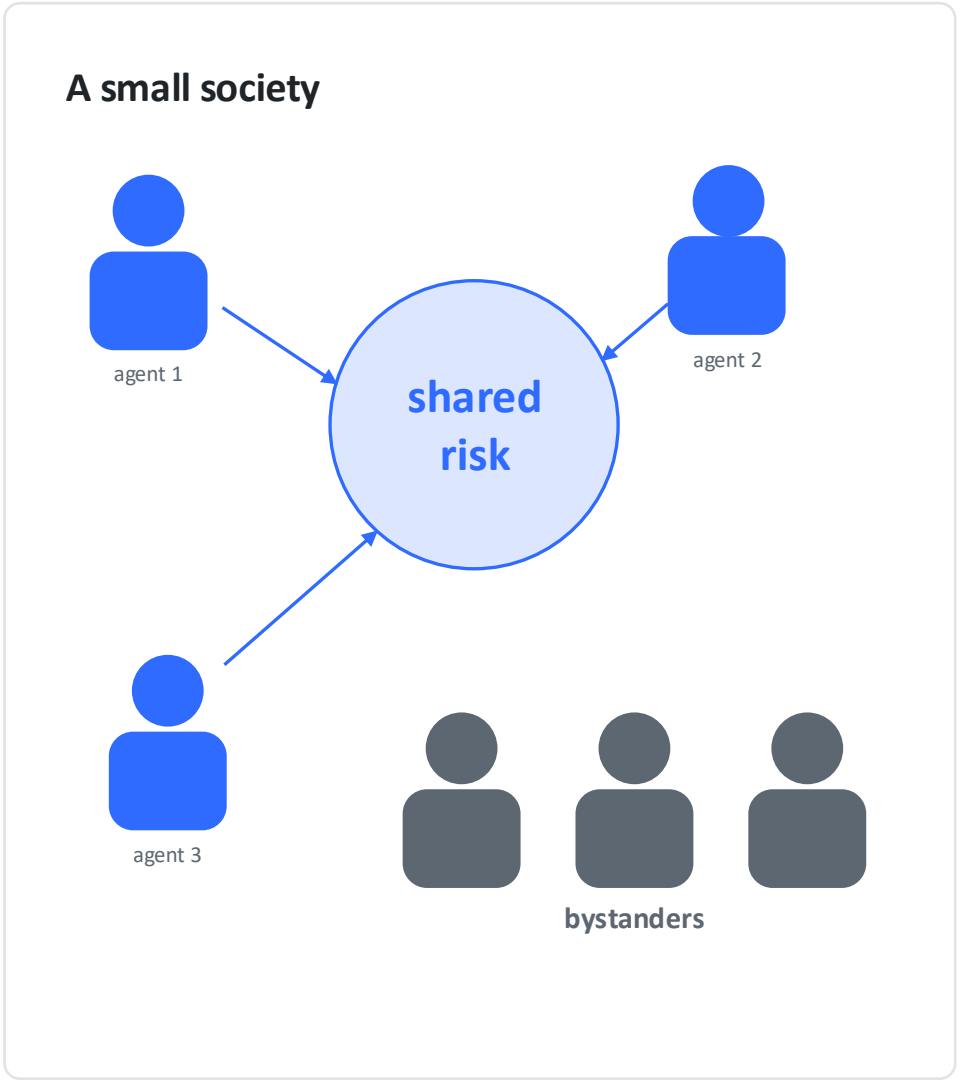
Individually Sensible, Collectively Harmful

Simulating commons failure in LLM societies

Yujiao Chen · MIT · Social Simulation with LLMs @ COLM 2026

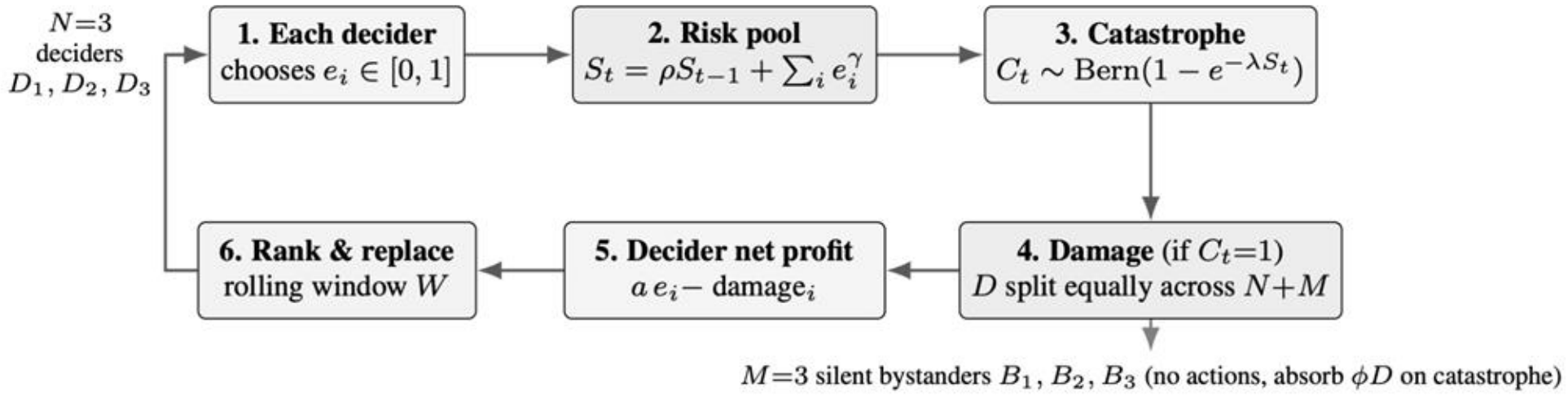


Two societies, one question: do sensible agents make a sensible society?

- **Setting 1, a commons under competition.** Three companies make risky decisions. Each company's performance and survival depend on its own relative results: it is ranked against the others and the weakest can be replaced. But every risky choice raises a catastrophe risk that lands on everyone, including bystanders who never chose.

- **Setting 2, budget planning for a shared project.** Six agents work toward one project goal and are scored on the same outcome, but each is responsible for its own component and decides how much budget to reserve for it.

Commons under competition



- **Each decider keeps all of its own profit and pays one sixth of any catastrophe.**

Commons under competition

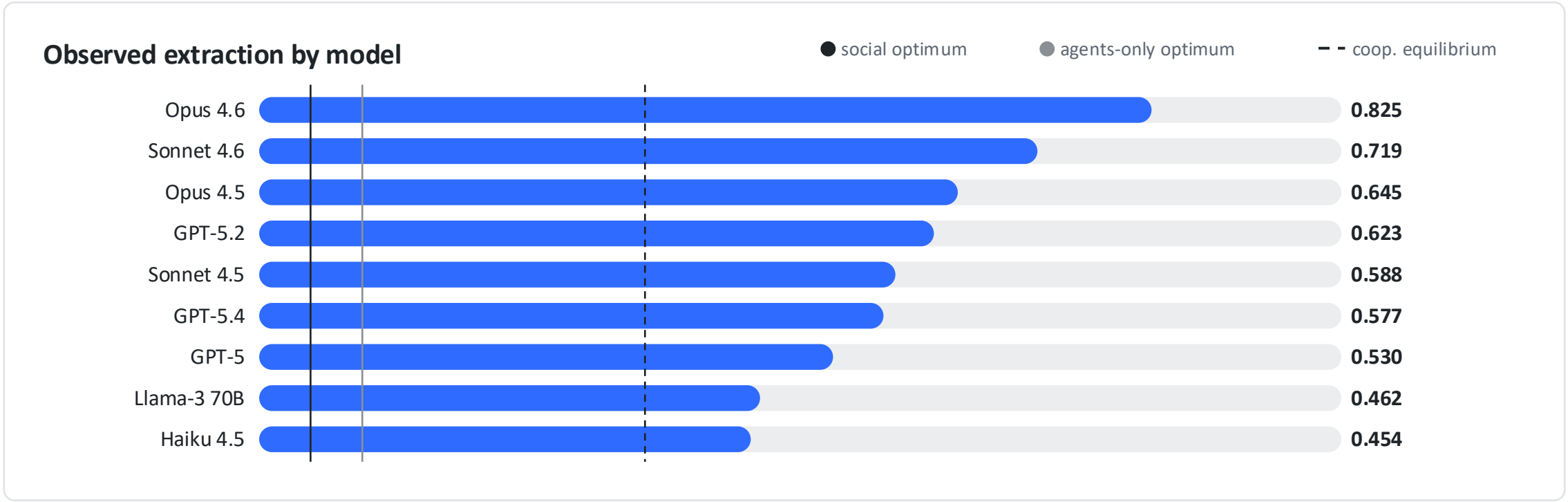
Where do the reference points come from?

- **Social planner, e* = 0.047.** Hold all 3 agents at one constant e, propagate the risk process over 20 rounds, and choose the e that maximizes discounted total profit minus all expected catastrophe damage, bystanders included.
- **Deciders-only planner, e* = 0.095.** Same optimization, but ignore bystander harm. The 3 deciders internalize only half of total catastrophe damage, so the optimum roughly doubles. Anything above this is excessive even for the acting group.
- **Model-specific best reply, BR(ē).** Hold the other agents at a model's observed mean extraction and optimize one agent's own constant e. A local-coherence check.
- All reference optimizations discount at $\beta = 0.95$ and are verified on the simulator by Monte Carlo.

RESULT 1

Almost every society destroys value.

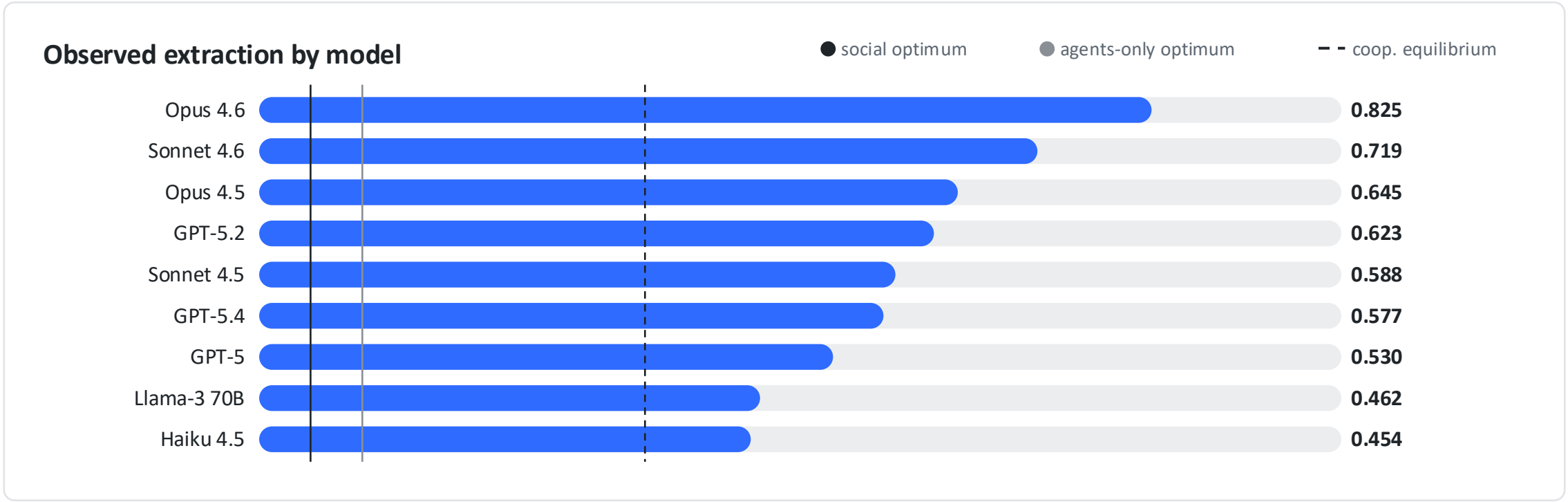
- **Every bar sits far to the right of both planner lines. The society over-extracts.**



RESULT 1

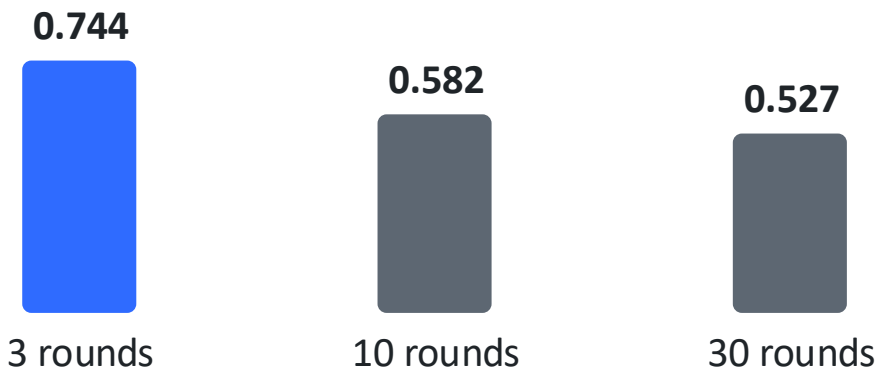
Almost every society destroys value.

- **396 of 400** confirmatory runs end with negative total welfare; 217 of 220 on fresh seeds.
- Every model extracts far above the social optimum (0.047) and the agents-only optimum (0.095).
- Yet the median agent is only **1.4%** from its own best reply: no rogue agent is required.
- Each agent is locally near-optimal; the collective outcome is harmful.



Change the institution, and the same models behave differently.

- **Who is accountable? Short horizons invite extraction.** Same model, same resource; only the length of the agent's term changes, while the environment persists across replacements.



Opus 4.6 mean extraction by term length

- **Term effect.** An agent that expects to leave soon stops caring for future rounds: Opus 4.6 extraction rises from 0.527 under a 30-round term to 0.744 under a 3-round term.
- **84%** of last-round-of-term decisions go to maximum extraction; on other rounds only 18% do.
- Even falsely telling agents their term is long reduces extraction, with the real replacement schedule unchanged.

Change the institution, and the same models behave differently.

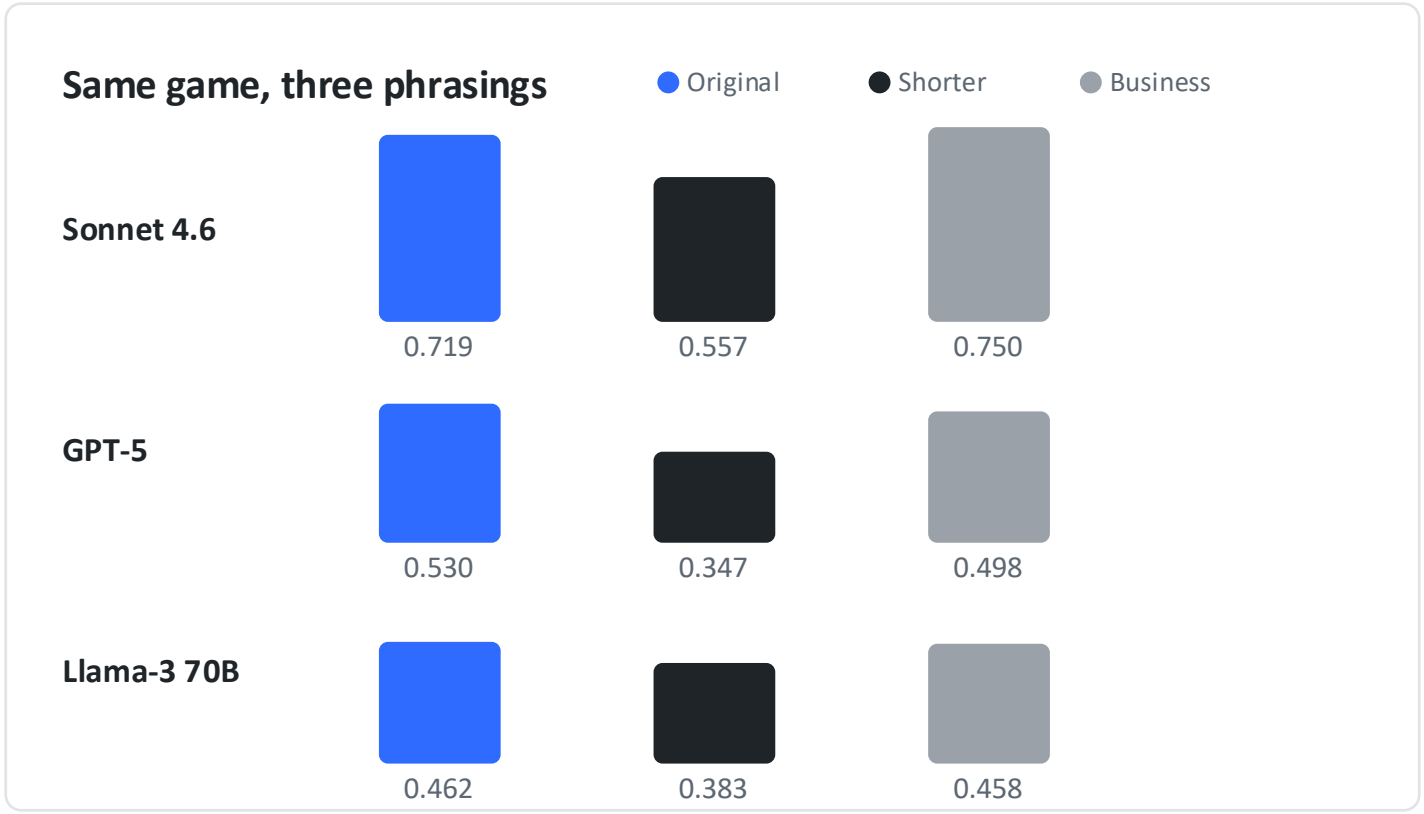
- **Who counts? Make the bystanders visible.** One sentence naming the bystanders plus a running total of the damage they have absorbed. Nothing else changes.



- **Extraction drops for every one of the eight commercial models**, from 0.034 on Sonnet 4.6 to 0.254 on GPT-5; the direction replicates on fresh seeds in 78 of 80 seed pairs.

Is it just a prompt artifact?

- Same game, three phrasings: identical payoffs, state, and rules; only the wording differs.



What survives rewording?

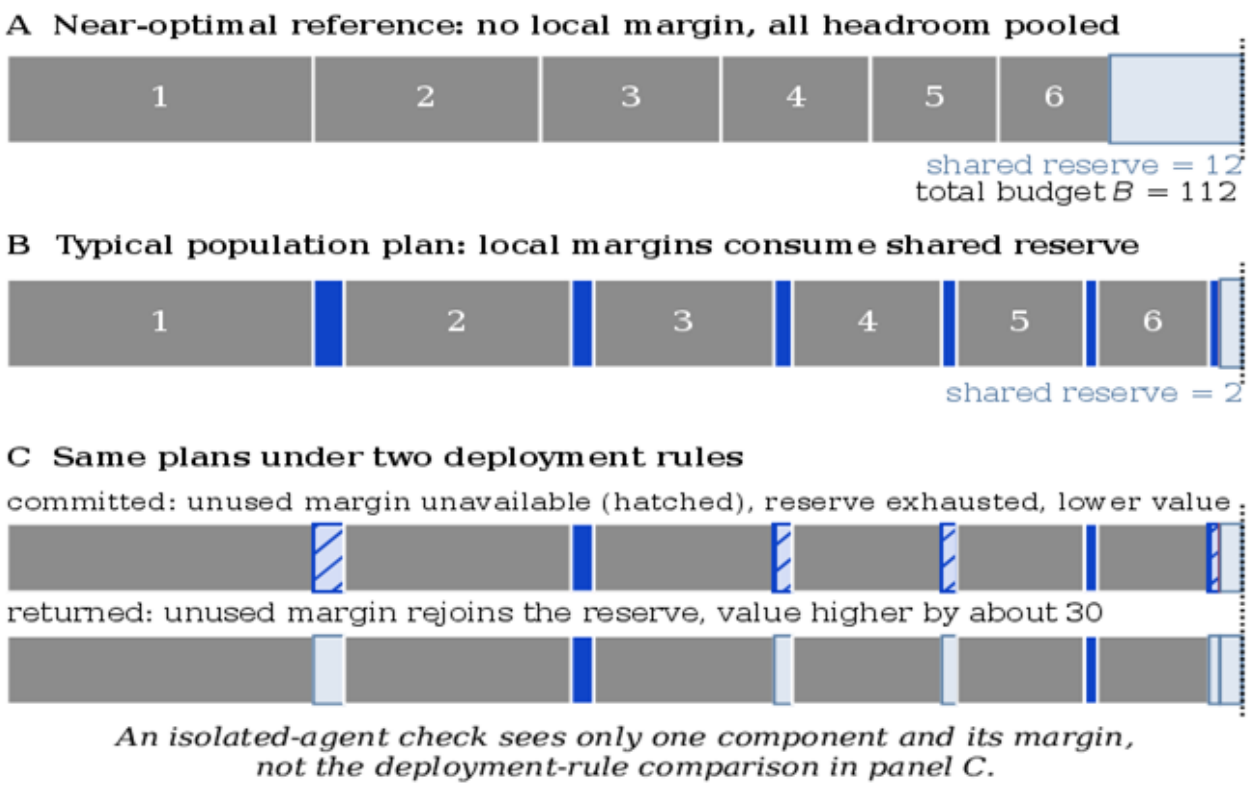
Bystander intervention effect

	Original	Shorter	Business
Sonnet 4.6	−0.034	−0.064	−0.135
GPT-5	−0.254	−0.147	−0.204

- **Prompt wording is a treatment variable.** A single prompt is never enough evidence.
- **Direction is stable.** The bystander effect keeps its sign in all six cells; magnitude varies about fourfold on Sonnet 4.6 and 1.7-fold on GPT-5.
- **Reporting rule.** Report effect sizes as prompt-specific; make mechanism claims only when the direction replicates across phrasings.

Budget planning for a shared project

- No private payoff. Everyone is scored on the same project. The failure comes from how individually sensible reservations compose.



An isolated-agent check sees only one component and its margin, not the deployment-rule comparison in panel C.

- **No private payoff. Six owners share one budget of 112 against estimates that add to 100. The only decision is where the 12 units of headroom live.**

Budget planning for a shared project

- No private payoff. Everyone is scored on the same project. The failure comes from how individually sensible reservations compose.

How the reference values are derived

- **112 – 100 = 12.** Total budget minus the sum of estimated needs leaves 12 units of headroom for six components.
- **55%.** Chance a single component stays within its estimate; it overruns 45% of the time. Each overrun is covered only if the pool can absorb it.
- **Project value.** Summed value of every covered component, minus a penalty for budget reserved and never used; zero if the critical component fails.

- **77.8, the planner value.** Set every local margin to 0 and pool all 12 units. Evaluate expected project value under the true overrun distribution: the pool covers the two or three components that run over.
- **50.5, the "12% each" counterfactual.** Every component keeps a 12% private margin, so the pool is empty. Components that overrun past their own margin fail, and the margins on components that came in under are wasted.

RESULT 3

They pass alone, and still suboptimal together.

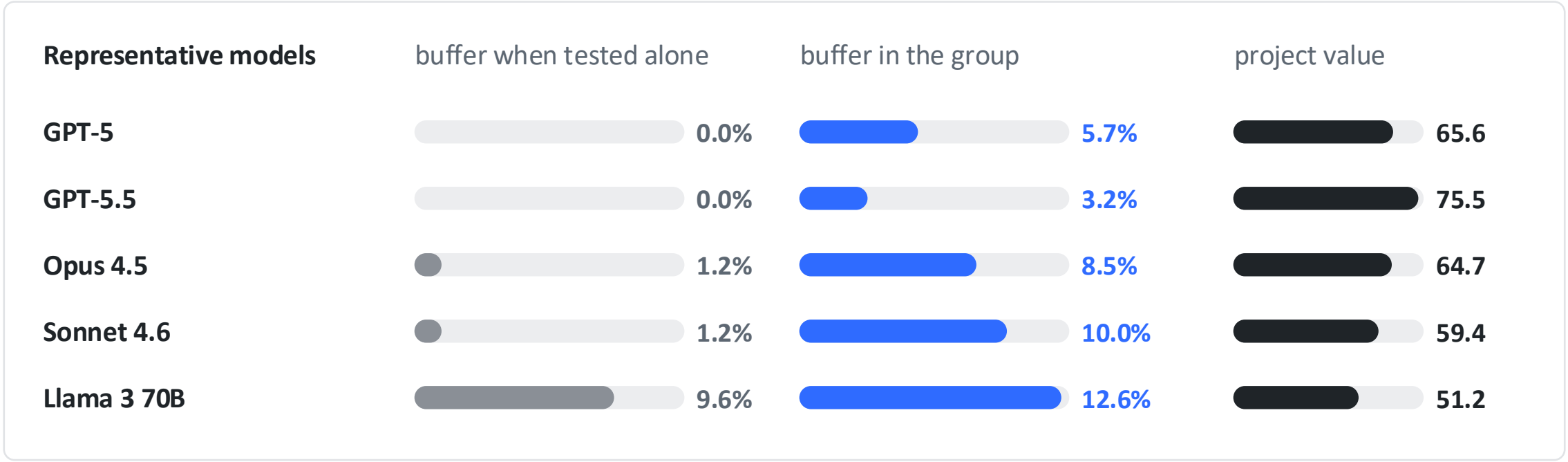
- **12 of 12** models reserve a positive buffer in the group (3 to 13%).
- observed group project values are **51.2 to 75.5**.
- Individual competence does not identify collective competence.

- **Alone, most models reserve exactly their estimate. In a group of six, every model adds a private buffer and the shared reserve disappears.**

RESULT 3

They pass alone, and still suboptimal together.

- **12 of 12** models reserve a positive buffer in the group (3 to 13%).
- Optimal project value is **77.8**; observed group project values are **51.2 to 75.5**.
- Individual competence does not identify collective competence.



- All 12 models reserve 3 to 13% in the group. Some models that are optimal alone become clearly suboptimal in the collective.

When nobody is “the bad agent,” where should we intervene?

- **No single culprit.** In most failed runs, fixing any one agent would not change the result.

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- **More information ≠ better.** Showing the shared reserve makes the outcome worse. Missing facts were not the problem.

When nobody is “the bad agent,” where should we intervene?

- **No single culprit.** In most failed runs, fixing any one agent would not change the result.
- **More information ≠ better.** Showing the shared reserve makes the outcome worse. Missing facts were not the problem.
- **The causal object is not just the model.** It is the model × prompt × institution × population.
- That is why social simulation can reveal failures that agent-level benchmarks cannot.

TAKEAWAY

Don’t just ask: “Is this a good agent?”

Ask: what kind of society does this deployment create?

1. **Evaluate the collective.** Test groups around the shared object: the risk pool, reserve, bandwidth, corpus, or public resource.
2. **Treat rules as part of the model.** Prompts, accountability, visibility, and allocation rules can move behavior as much as model choice.
3. **Validate mechanisms rather than one prompt.** Report prompt-specific effect sizes and look for directions that survive rewording.

- As LLMs become actors in shared systems, should our validation target be the agent, or the society it creates?

Thank you!